

Rent or Toxicity? Strategic Exploitability of Learning Market Makers with Public Wallet Information (Hyperliquid)

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Motivation

Market makers (MMs) post bids and asks and earn the spread from impatient traders, while losing to *informed* traders whose orders predict the next price move (adverse selection). Increasingly, MMs are learning algorithms. Three facts make this a trading-firm problem, not only a regulator’s. (i) Independent learning MMs drift to spreads above the competitive level without communicating [7, 1]. (ii) A spread above competitive is *rent* an entrant can capture by undercutting, but the same wide spread is also the correct response to toxic flow: identical observable, opposite action. (iii) Optiver reports that current LLM traders name adverse selection yet trade at negative expected value and fail to model how others react [16]. The skill a desk wants—read the tape, decide whether competitors are soft or the flow is poisonous, and quote accordingly without becoming the exploited party—has no quantitative benchmark.

Central claim. Learning market makers earn predictable supra-competitive rents; the regularity that sustains those rents is itself a strategic vulnerability; public market and wallet information lets an adaptive entrant exploit it; robust quoting can partially restore the profit–resilience frontier. The pieces exist separately: learned rents [1, 7]; strategic exploitation of learners in abstract games [10, 4]; on Hyperliquid, persistent wallet histories carry adverse-selection information [2] and disclosed flow changes impact [3]. Nobody has asked whether learned rents survive a best-responding rival, or turned wallet-level predictability into executable quoting under competition—the gap [2] leaves open.

We study this on **Hyperliquid**, an on-chain central limit order book (CLOB) for perpetual futures with billions of USD in daily turnover. Every committed order, cancel, rejection, and fill carries a persistent *pseudonymous wallet address*, with both counterparties and post-trade inventory recorded [14]. Hyperliquid is our empirical laboratory and calibration source; the venue mechanics in Section 1 are treatment arms, not individual novelty claims.

1 Setting and Model

Operational setting. One perpetual contract on a Hyperliquid-style CLOB. Time is indexed by committed blocks $t = 1, \dots, T$, a tractable event-time approximation consistent with the venue’s atomic state-update structure (blocks arrive on an irregular clock, median ≈ 68 ms, and order priority is continuous in the priority fee within the block logic [15]). An episode is $T = 100$ blocks; learning runs across 10^4 – 10^5 episodes. We write m_t for the mid-price, θ_t for tick size, and V for the fundamental (the composite index Hyperliquid marks against).

Agents.

- (a) *Incumbent MMs* $i \in \{1, \dots, N\}$, $N \in \{2, 3, 5\}$. Learners (Q-learning, Hedge/no-regret, PPO). Action $a_t^i = (\delta_t^{b,i}, \delta_t^{a,i}, q_t^i)$: bid/ask offsets from m_t on the tick grid, and displayed size. Objective: $\max \mathbb{E} \sum_t \gamma^t [\pi_t^i - \phi(I_t^i)^2]$ where π_t^i is trading PnL net of fees and I_t^i inventory. Resource constraints: inventory cap $|I_t^i| \leq \bar{I}$ and a *request budget* B_t^i (Section 1).
- (b) *Strategic entrant (the trader)*. One MM who observes the public tape, infers whether the spread is rent or toxicity, and best-responds. The protagonist; later evaluated as the “robust quoter.”
- (c) *Takers*. One arrival per block: informed with probability α (trades in the direction of $V - m_t$; on Hyperliquid this is empirically latency arbitrage against Binance/OKX mid-prices), else a liquidity trader with private valuation $m_t + L$, $L \sim G$. α is the *toxicity* dial. In the identity treatment each taker carries a persistent wallet id whose past fills are public *after commitment*: a maker at t sees wallet histories through $t - 1$, never the identity of the order about to hit it.
- (d) *Predatory taker*. An informed trader who chooses *timing* and *size* against incumbents’ learning dynamics (provoke a price war, then trade into it; strike when a competitor’s budget is exhausted).

- (e) *Nonstrategic disruption: liquidation flow.* With hazard $\lambda(\sigma_t)$ a forced market order of size $S \sim F$ arrives, routed to the book, then to a backstop vault [15]. Stress case: the 10 Oct 2025 cascade, with $\approx \$10.3\text{B}$ of 24-hour liquidations attributed to Hyperliquid and multi-venue spreads $\sim 30\times$ wider in the worst 40 minutes [17]; we compute Hyperliquid’s own spread/depth path from [14].

State and information. $s_t = (\mathcal{B}_t, m_t, \mu_t, \{I_t^i, B_t^i\}_i, \sigma_t, z_t)$: the visible book, public belief $\mu_t = \mathbb{E}[V \mid \text{tape}]$ updated by Bayes’ rule after each print, inventories, request budgets, a volatility state driving λ , and z_t = public flow features (wallet histories in the identity treatment). Information sets are nested, $\mathcal{F}_t^{\text{anon}} \subset \mathcal{F}_t^{\text{id}} \subset \mathcal{F}_t^{\text{oracle}}$ (no ids / committed wallet histories / the latent flow type, an infeasible upper bound), which turns “identity” into a value-of-information experiment. Fills follow price–time priority; the taker hits the best quote; ties resolved by queue position. **Payoffs.** Per fill $\pi_t^i = (p - V)$ when selling, $(V - p)$ when buying, plus maker rebate r (negative fee at high-volume tiers) minus taker fee f for aggressive orders. We report quoted/effective/realized spread (markouts at 1, 10, 60 blocks), taker welfare, $|\mu_t - V|$, MM PnL mean and CVaR, and the best-response loss defined under RQ2. **Sequence.** Quote $\rightarrow$ taker/liquidation $\rightarrow$ match $\rightarrow$ print, belief and budget update $\rightarrow$ learn; at episode end V is revealed. **Benchmarks (exact).** Competitive Glosten–Milgrom quotes on the grid ($\text{ask} = \lceil \mathbb{E}[V \mid \text{buy}, s_t] \rceil_\theta$); joint-monopoly quotes; collusion index $\Delta = (\bar{\pi} - \pi^C)/(\pi^M - \pi^C)$ [11]. **Simplifications.** One asset, unit taker size, risk-neutral MMs, exogenous taker mix, no cross-venue hedging, calibrated (not endogenous) liquidation sizes.

Why Hyperliquid: treatment arms and calibration

Prior collusion/RL market-making papers use stylized dealer games or anonymized equity data. Hyperliquid supplies mechanics that serve as treatment arms in the simulator and as calibration targets in the data:

1. **Persistent wallet identity.** Every committed order carries a pseudonymous address (a principal may run several). MMs in an *anonymous* equity book pay to signal identity and avoid sniping each other [8]; Hyperliquid wallet toxicity persists across ten-day windows (rank corr. 0.52) and improves return forecasts out of sample [2]. We measure whether MM wallets avoid each other, whether the best-quoter identity alternates (our learners’ turn-taking signature), and whether entrants are punished. Anonymous vs. pseudonymous is a design arm.
2. **Request budgets.** Address-based limits grant one action per USDC traded plus a buffer; a throttled address gets one request per 10 s; cancels get headroom; limits are per user [15]. Quoting capacity is *earned by filling*, $B_{t+1}^i = B_t^i + g(\text{fills}) - c(\text{updates})$. Nobody can decrement another wallet’s counter, so the attack is *induced self-exhaustion*: stimuli (e.g. flickering quotes, which do elicit responses [5]) that make a reactive rival’s own policy burn its budget; defences are hysteresis, minimum quote lifetime, a reserve.
3. **Priced latency.** Cancels precede executable orders; for aggressive orders a priority fee buys $\approx 45\text{ms/bp}$ up to 8 bp, for resting orders it buys queue position within a 400 ms window [15]. For a maker, priority is costly *queue-position control* (action component ρ_t^i); the arms race of [13] has an explicit price schedule.
4. **Tick grid.** At most five significant figures subject to asset-specific decimal limits, integers always allowed [15]; we compute the effective grid from the venue rule per asset and price region. Tick coarseness governs learned supra-competitive pricing [9]; power-of-ten crossings are natural experiments.
5. **HLP-inspired fixed benchmark.** The protocol vault runs several undisclosed strategies with on-chain positions [15]; a fixed-spread/inventory-skew quoter *calibrated to observable HLP behaviour* is the rubric’s fixed-policy agent.
6. **Data.** Open Level-4 data: every order lifecycle for BTC/ETH/SOL in Dec 2025, all trades Oct 2025–Jan 2026 [14]; the public feed also tags every print with both wallets. Calibrates α , G , F , λ and validates simulated reaction functions against real MM wallets.

2 Relevant Literature

- Colliard, Foucault & Lovo [1] (RFS 2026): Q-learning MMs in Glosten–Milgrom adapt to adverse selection but fail to learn competitive pricing because undercutting feedback is noisy; rents are larger when adverse selection is weak. *We take* their environment and exact benchmarks as baseline and repli-

cation target; our contribution begins *after* the rents emerge: learned policy $\rightarrow$ observable behavioural signature $\rightarrow$ strategic best response.

- **Zhai [2] (2026):** on 17.1B Hyperliquid L4 messages, wallet-level toxicity is persistent and wallet-history features improve one-second return forecasts out of sample; turning this into profit “would require a dynamic execution and market-making model accounting for latency, queue priority, fill probabilities, fees, and inventory” and is left open. *We take* the validated signal and the terminology (wallet identity, not trader identity). *We differ* by asking whether the signal survives competition from adaptive rivals and becomes executable.
- **Deng, Schneider & Sivan [10] (NeurIPS 2019) and Brown & MacKay [4] (2026):** the first proves no-regret learners can be steered by a strategic opponent; the second shows a fast, persistent reaction rule can coerce supra-competitive outcomes even against a learning rival. *We take* the steering attacker and the view of learned punishment as algorithmic commitment. *We differ* by instantiating both in a market-making game with adverse selection and asking whether the regularity that sustains rents is what makes them exploitable. Supporting: [7, 8, 3, 5, 6, 12].
- **How decisions are made today.** Hyperliquid’s public documentation specifies matching, rate limits, priority fees and liquidation routing [15]; Optiver’s blog describes the reasoning MMs expect of traders and where models fall short [16].

3 Research Questions

RQ1 (Executable value of information). What is an entrant’s profit under nested information, $\pi^{\text{anon}} \leq \pi^{\text{id}} \leq \pi^{\text{opp}} \leq \pi^{\text{oracle}}$: book and flow only; plus lagged wallet-level toxicity; plus an estimated incumbent reaction function; plus the latent flow type (upper bound)? *Benchmarks:* competitive-GM quoting and a fixed-spread/inventory-skew heuristic. *Measure* net PnL per unit volume, realized markout, market share, inventory risk and CVaR_{0.05} across $\alpha \in \{0.1, 0.3, 0.5\}$, $N \in \{2, 3, 5\}$. *Prediction:* rent is highest where toxicity is lowest [1], so a spread-width heuristic enters the wrong markets. This is the dynamic-execution question [2] leaves open.

RQ2 (Does learned rent create strategic vulnerability?) Characterise each trained incumbent population by its rent $C(\pi) = (\Pi(\pi) - \Pi^C) / (\Pi^M - \Pi^C)$; freeze it; compute the best-response loss $L_{\text{BR}}(\pi) = J(\pi; \text{nominal}) - \min_{\pi^A} J(\pi; \pi^A)$ (the market is general-sum, so incumbent loss, not attacker gain, is the exploitability statistic). *Hypothesis:* $\partial L_{\text{BR}} / \partial C > 0$ —the regularity required to sustain rents makes the policy easier to manipulate. *Attackers*, escalating: one-tick undercutter; a strategic learner exploiting reaction rules [10]; induced request exhaustion; a predatory taker; held-out best responses (PSRO), with $\mathcal{A}_{\text{train}} \cap \mathcal{A}_{\text{test}} \approx \emptyset$. *Incumbents:* Q-learning, no-regret/Hedge, one deep-RL method, HLP-inspired fixed.

RQ3 (Defences; extension). Quoter levers: randomisation, update hysteresis, budget reserve, identity-aware toxicity control, inventory limits. Venue levers: anonymity, tick grid, request allowance, priority mechanism. *Report* the frontier $(\mathbb{E}[\Pi_{\text{nominal}}], -\text{CVaR}_{0.05}(\Pi_{\text{attack}}))$ with quoted spread, fill rate and taker cost, so “robust” cannot mean “withdrew”. *Empirical anchor:* from [14], $P(\text{maker requotes within } h \mid \text{rival undercuts by a tick})$ for active MM wallets vs. the learned agents’ reaction functions; wallet features use prior observations only.

4 Plan and Feedback

By Workshop 1 (deliberately narrow). (i) Two tabular-Q incumbents in Glosten–Milgrom replicating [1] qualitatively; (ii) exact benchmarks, $C(\pi)$, a collusion certificate; (iii) one markout-rule entrant vs. competitive GM at $N = 2$, three α ; (iv) BTC spread, markout, arrival and toxicity statistics on a manageable Hyperliquid sample. Deep RL, PSRO, queues, cascades, priced priority and request exhaustion are later-stage work. **Rigor.** Common random numbers across policies; confidence intervals over seeds; CVaR with mean PnL; train at α , test at α' ; held-out attackers. **Target.** arXiv preprint by mid-December as a milestone toward peer review.

Feedback requested. (1) Stylized Glosten–Milgrom (exact benchmarks) or queue-based LOB (realistic, approximate)? We lean stylized plus request budgets and priced latency. (2) Is induced request exhaustion in scope—legal on the venue, adversarial in spirit? (3) We read the 10–100 node guidance as met by N incumbents, entrant, takers and attackers on one book, and would rather not add a multi-asset extension that raises dimensionality without sharpening the claim—acceptable?

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